

OFFICIAL ARCHITECTURE SPECIFICATION

IvoxStack Platform

Comprehensive Technical Specification, UI/UX Design System, Frontend Architecture & Backend Blueprint

A production-grade, end-to-end digital solutions agency platform engineered with React 18, Vite, TypeScript, Tailwind CSS, Supabase PostgreSQL, and Netlify serverless automation.

PLATFORM NAME	CORE TAGLINE
IvoxStack (ivoxstack.com)	Digital Solutions Built for Business Growth
ARCHITECTURE MODEL	GENERATED DATE
Decoupled Frontend SPA + Serverless Backend + Supabase	September 2026

1. UI/UX Design System & Aesthetics

The platform strictly avoids arbitrary color blends on typography and uses a unified, high-contrast, premium aesthetic built around solid slate typography, vibrant call-to-action triggers, and responsive interactive surfaces.

1.1 Core Color Palette

ROLE / COMPONENT	HEX CODE	TAILWIND TOKEN	PURPOSE & VISUAL HIERARCHY
Primary CTA Button	 #f95700	<code>bg-[#f95700]</code>	High-conversion action trigger for consultations, inquiries & proposals.
WhatsApp Direct	 #00a884	<code>bg-[#00a884]</code>	Official WhatsApp Business green for instant chats & pre-filled inquiries.
Solid Primary Heading	 #020617 / #0f172a	<code>text-slate-950/900</code>	Single professional solid color for all H1, H2, and H3 headers (Zero gradients).
Brand Blue Accent	 #0066cc	<code>text-[#0066cc]</code>	Technical trust badge, audit triggers, and active navigation indicators.
Card Surfaces	 #ffffff / #f8fafc	<code>bg-white / bg-slate-50</code>	Clean container contrast with subtle 1px slate borders (#e2e8f0).

1.2 Typography & Geometrics

Typography System

Inter (Google Fonts) with fallback to system sans-serif.

- Headings: font-black (900) & tracking-tight
- Body Text: font-normal (400) / font-medium (500)
- Badges & Micro-tags: font-bold (700) uppercase

Border Radii & Micro-Curves

All interactive cards, modals, and input fields are constrained to **12px to 13px** curves (rounded-2xl & rounded-3xl). Avoids childish over-rounded capsules while preserving a smooth modern feel.

2. Frontend Architecture & Technology Stack

The client is structured as an ultra-fast Single Page Application (SPA) powered by **Vite 6** and **React 18** with complete static type safety via **TypeScript 5**.

2.1 Technologies & Libraries

LIBRARY / FRAMEWORK	VERSION	ROLE IN APPLICATION
React	^18.3.1	Component-based UI state rendering, virtual DOM reconciliation, and hooks.
Vite	^6.1.0	Ultra-fast ES-module dev server and lightning-fast Rollup production bundling (~2.7s).
TypeScript	^5.7.3	Strict type contracts across CRM pipelines, state models, and props.
Tailwind CSS	^3.4.17	Utility-first CSS engine customized with custom border-radii and color tokens.
React Router DOM	^6.28.2	Client-side routing, URL query parameter synchronization, and hash scrolls.
Lucide React	^0.475.0	Crisp SVG vector iconography across services, industries, and CRM dashboard.

2.2 Public Website Structure (src/pages/)

- **HomePage** (HomePage.tsx): Hero rotating orbit, starting prices grid, 15 core services, 4-step framework, portfolio showcases, 16 industry vertical cards, embedded cost calculator, and digital audit callout.
- **Services Page** (ServicesPage.tsx): 15 end-to-end capabilities with hash navigation anchors (#slug).
- **Pricing Page** (PricingPage.tsx): 11 dynamic categories synced with URL search params (/pricing?category=creative).
- **Portfolio Page** (PortfolioPage.tsx): Filterable enterprise decks, live project links, and embedded PDF previewer.
- **Case Studies** (CaseStudiesPage.tsx): Real client business metrics, ROAS reductions, and lead volume scaling.
- **Interactive Cost Calculator** (CalculatorPage.tsx): Live cost estimation engine with direct WhatsApp pre-filled export.
- **Digital Audit** (DigitalAuditPage.tsx): 4-step audit request engine for SEO, speed, Meta Pixel, and ad copy.
- **Legal Pages** (LegalPages.tsx): Privacy Policy, Terms of Service, Refund Policy, Cookie Policy, Revision Policy.

3. Backend & Database Architecture

The backend is separated into two dedicated sub-layers: a cloud relational PostgreSQL database managed via **Supabase**, and a serverless API function layer hosted on **Netlify Functions**.

3.1 Database Layer (backend/database/)

Configured with standard SQL migrations containing relational tables, indexes, and Row Level Security (RLS) policies:

SQL MIGRATION FILE	CONTENTS & DATABASE SCOPE
001_initial_schema.sql	Defines core tables: leads (CRM inquiries), clients, projects, invoices, services, pricing_plans, portfolio_items, activity_logs, and platform_settings.
002_rls.sql	Row Level Security policies restricting public write access to lead capture while requiring authenticated sessions for admin analytics and CMS mutations.
003_seed_data.sql	Pre-populates starting prices, turnkey bundles, 15 core services, initial test leads, and official business configuration.

3.2 Serverless Functions Layer (backend/functions/)

create-lead.ts

Validates incoming customer inquiries, runs anti-spam honeypot checks, verifies elapsed submission time, and persists the lead to Supabase.

admin-stats.ts

Computes key business intelligence: Total Leads, Monthly Recurring Revenue (MRR), Active Projects, and conversion velocity.

export-leads.ts

Generates clean, sanitized CSV downloads of pipeline leads for external CRM synchronization and sales follow-ups.

backup-data.ts & restore-data.ts

Provides JSON cloud backup snapshots and disaster recovery restoration triggers directly from the Operations portal.

4. Operations Admin Portal & CRM (`src/admin/`)

IvoxStack includes a built-in enterprise management portal located at `/operations/login`. It operates with a dual data model: live Supabase cloud sync when credentials are provided, or instant zero-setup local storage fallback.

ADMIN MODULE	FILE PATH	KEY FUNCTIONAL CAPABILITIES
Dashboard	<code>src/admin/Dashboard.tsx</code>	Total Revenue, Active Projects, Pipeline Lead Count, and Recent Activity stream.
Leads & Sales CRM	<code>src/admin/LeadsCRM.tsx</code>	Filter by status (NEW, CONTACTED, QUALIFIED, WON, LOST), budget filter & search.
Client Directory	<code>src/admin/Clients.tsx</code>	Customer database, total spend, status, and associated ongoing deliverables.
Project Delivery	<code>src/admin/Projects.tsx</code>	Tracks deliverables, milestone stages (Draft, Review, Delivery), and due dates.
Invoices & Payments	<code>src/admin/Invoices.tsx</code>	Generates billing records, payment status tracking (Paid, Unpaid, Overdue).
Services CMS	<code>src/admin/ServicesCMS.tsx</code>	Create, edit, toggle visibility, and adjust starting rates for all 15 services.
Pricing Catalog CMS	<code>src/admin/PricingCMS.tsx</code>	Manage prices, feature checklists, and "Most Popular" tags across all packages.
Activity & Audit Logs	<code>src/admin/ActivityLogs.tsx</code>	Timestamped operational log recording logins, lead mutations, and CMS edits.
Platform Settings	<code>src/admin/Settings.tsx</code>	Global business WhatsApp number, phone, email, address, and maintenance mode toggle.

4.1 Analytics & Telemetry (`src/lib/analytics.ts`)

Automatically reads and captures standard marketing query parameters (`utm_source`, `utm_medium`, `utm_campaign`, `utm_term`, `utm_content`). These parameters are stored in `sessionStorage` and automatically attached to lead submissions and WhatsApp URL click triggers.

5. Clean Project Structure & Summary

The project has been cleaned of all legacy mockups and test files, presenting an intuitive folder layout:

```
IvoxStack/  
├── backend/           # Database schemas & serverless functions  
│   ├── database/     # 001_initial_schema.sql, 002_rls.sql, 003_seed_data.sql  
│   └── functions/    # Netlify API endpoints (create-lead, admin-stats, etc.)  
├── public/           # Production assets (ivoxstack-icon-transparent.png, nva-infrac  
├── src/              # Client Application  
│   ├── admin/        # 15 Operations Portal & CRM pages  
│   ├── components/   # 10 Reusable UI elements (BrandLogo, Header, Footer, Modals)  
│   ├── lib/          # State store, Supabase client, Analytics & Utilities  
│   ├── pages/        # Public pages (Home, Services, Pricing, Legal)  
│   ├── types/        # TypeScript centralized interfaces  
│   ├── App.tsx       # Routing shell  
│   ├── index.css     # Tailored design tokens  
│   └── main.tsx      # React root entry  
├── netlify.toml      # Serverless deployment configuration  
├── package.json      # Project manifest (ivoxstack)  
└── README.md         # Complete beginner-friendly guide
```

✓ **Build Verified:** Production bundle compiled with zero type errors (Built in 2.70s with Vite).